PRANALI SAXENA

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SUMMARY

M.S. Business Analytics & AI candidate with industry experience at Capgemini and Deloitte engineering AI/ML model management workflows, RAG-powered LLM systems & production ETL pipelines. Proficient in Python, SQL, Scikit-learn, LangChain, and RESTful APIs. Seeking 2026 Data Science / AI-ML / GenAI internships in the USA.

EDUCATION

University of Texas at Dallas, Richardson, Texas — **Aug 2025 - May 2027**

Master of Science in Business Analytics and Artificial Intelligence ***(Dean's Excellence Scholarship)***

Coursework: Database Foundations, Prescriptive Analysis, Applied ML, Business Analytics with R, Advance Statistics, Organizing Platforms for BA.

Manipal University, India — **Jul 2020 - May 2024**

Bachelor of Technology in Electronics and Telecommunication Engineering ***(Dean's List Award)***

Coursework: C, DBMS, Data Structures and Algorithms, Operating Systems, Big Data Analytics, AI, Cloud Computing, Cyber Security, IoT, OOP.

PROFESSIONAL EXPERIENCE

Capgemini | Gurgaon, India | AI-ML Analyst — **Sept 2024 - Aug 2025**

- **Designed** and **automated AI/ML model management workflows** for the NON-RT RIC (Non-Real-Time RAN Intelligent Controller) compliant with O-RAN Alliance specifications, leveraging **Net-Anticipate** for ML-driven network optimization and building **Python** and **Golang** solutions.
- **Consolidated multi-source O-RAN datasets (PostgreSQL, MongoDB, Influx DB)** via Python and SQL ETL pipelines to power KPI dashboards, built an LLM-assisted RAG & LangChain analytics assistant to automate EDA & surface anomalies, cutting analyst query turnaround by **27%.**
- **Administered Linux / MobaXterm remote server workflows** & enforced API interoperability across 4 heterogeneous databases, reducing error rate.

Secure Meters Limited | Udaipur, India | Business Intelligence Intern — **Jan 2024 - Sept 2024**

- Applied feature engineering, anomaly detection on **5M+ smart meter records** (UK, Ireland, Germany) using Python, SQL to surface energy consumption patterns & grid inefficiencies, delivered load-balancing recommendations that **directly accelerated cross-regional operational planning.**
- **Built SQL & Python demand forecasting pipelines** to detect peak load and usage variance, constructed KPI dashboards (grid efficiency, usage variance), cutting time-to-insight by **15%** and improving demand forecasting accuracy across three countries.

Maruti Suzuki India Limited | Gurgaon, India | Data Analyst Intern — **Jul 2023 - Dec 2023**

- Executed a **data-driven POC** for microphone placement using **multivariate analysis, A/B testing, statistical signal evaluation** on **10K+ in-vehicle audio data points**, quantified **12% SNR improvement**, **18% reduction** in microphone interference across 3 vehicle models through hypothesis testing.
- Translated **root-cause statistical findings** into structured product design recommendations; influenced front/rear seat microphone placement decisions adopted across 3 vehicle models, improving in-vehicle communication clarity in line with automotive engineering standards.

Deloitte | Gurgaon, India | Data Analyst Intern — **May 2023 - Jul 2023**

- Built **customer churn prediction pipeline** on **1M+ CRM records**, performed EDA, feature engineering, data preprocessing via SQL & Python, trained **Logistic Regression (baseline)**, **Random Forest** and **XGBoost** with cross-validated hyperparameter tuning, achieving **ROC-AUC 0.84** and **F1-score 0.79.**
- Applied **SHAP-based feature importance** to identify top churn drivers, visualized **customer segmentation** and KPI trends via Tableau/Power BI dashboards, delivering **data-driven retention recommendations** that improved client **decision making efficiency** by **12%**.
- **Certified Salesforce Developer** (SmartInternz × Salesforce × Deloitte) - Apex, Lightning Web Components, Flow Automation, VS Code CLI.

PROJECTS

NYC FHVHV Tip Prediction - Two Stage ML Pipeline — *Python · LightGBM · CatBoost · Neural Networks · MLP · Feature Engineering*

- Built a two-stage ML pipeline (classifier - regressor) on **20.94M NYC trip records**, applied MNAR-aware imputation, temporal train/test split, and target encoding across 265 high-cardinality zones using **LightGBM, CatBoost, and MLP** to predict tip amounts for driver earnings analytics.
- Achieved MAE $1.72 (3.4% improvement over baseline) via cross-validated hyperparameter tuning and SHAP-driven feature selection, improving demand forecasting accuracy for a platform scale dataset.

Synapse AI - WEHack UTD '26 × CBRE — *LangChain · FastAPI · Python · Multi-Agent AI · GenAI*

- Architected CBRE Asset Brain, a **5-agent LLM orchestration system** integrating Isolation Forest for anomaly detection, XGBoost classifier for tenant risk scoring on 40K+ structured lease records, and RAG-powered lease signal tracking, served via 11 FastAPI REST endpoints with a 3-tier memory architecture (episodic, semantic, procedural).
- Improved tenant risk classification **F1-score** from **0.41 to 0.57** and overall decision accuracy 38% to 53% over 8 weeks via memory accumulation, delivering 3× faster anomaly resolution and 60% reduction in manual review time.

TECHNICAL SKILLS

Languages & Libraries: Python, SQL, R, Golang, VBA, Power Query M, Pandas, NumPy, Matplotlib, Seaborn, Rust, HTML, CSS.

AI / ML & GenAI: Scikit-learn, LightGBM, CatBoost, XGBoost, TensorFlow, Keras, SMOTE, Feature Engineering, Ensemble Methods, LangChain, RAG, Prompt Engineering, OpenAI, API, Agentic AI, Multi-Agent Orchestration, LLM, NLP.

Data Engineering & Cloud: ETL Pipelines, Apache Spark, Kafka, Airflow, DBT, Snowflake, Azure Data Lake, AWS S3, Microsoft Fabric, Alteryx, RESTful APIs, FastAPI, Data Modeling, Data Warehousing, Kubernetes, Docker, CI/CD.

BI & Databases: Power BI (DAX), Tableau, Metabase, KPI Dashboard Development, Data Storytelling, PostgreSQL, MySQL, MongoDB, Snowflake, Influx DB

Tools & Concepts: Git, Linux, Jupyter, Docker (basic), MS Excel, Jira, EDA, A/B Testing, Hypothesis Testing, Statistical Inference, Anomaly Detection, Time-Series, Predictive Modelling, MLOps, Data Wrangling, O-RAN.

CERTIFICATIONS (2025-2026)

IBM - Deep Learning with PyTorch, Keras & TensorFlow, **SAP** - AI Core Services on SAP BTP, **Amazon** - Generative AI and AIML Fundamentals.